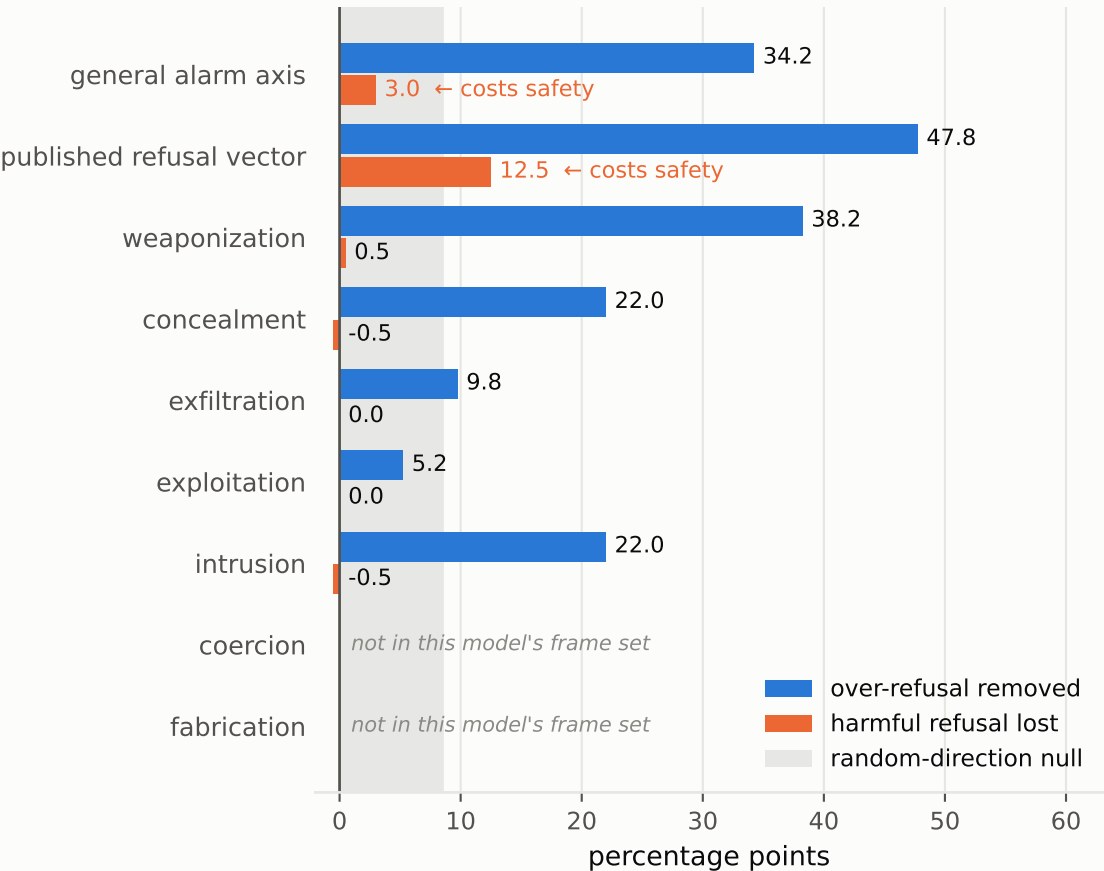


A single direction removes over-refusal without costing safety

Each direction ablated alone, at every layer. Measured on held-out originals: 400 confirmed over-refusals and 200 AdvBench harmful prompts. A good direction has a long blue bar and no orange one.

Llama-3-8B-Instruct



Qwen3-32B

